Banach property (T) by finite averaging

The infinite group $\mathrm{SL}_3(\mathbb{F}_q[t])$, in two cases

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Abstract

We reorganize the polynomial-ring proof around one finitely supported averaging operator A. Given any vector x and any invariant vector y, either an upper- or lower-triangular average moves $x - y$ noticeably, in which case uniform convexity makes Ax closer to the same y, or neither does, in which case the entire orbit of x is small and supplies a new invariant vector y'. The result is $\|Ax - y'\| \le \theta \|x - y\|$ for a fixed $\theta < 1$. We retain the six-factor Heisenberg interpolation from the draft, but use finite comparisons rather than limits of averaging operators. The sole completeness step is an elementary small-orbit lemma, proved here. Nica's bounded elementary generation theorem is the only external algebraic input. No finite-graph applications are included.

1 The statement and the proof's two ingredients

Fix a finite field $\mathbb{F}_q$ and put

$$\Gamma = \mathrm{SL}_3(\mathbb{F}_q[t]), \qquad e_{ij}(p) = I + pE_{ij}.$$

A linear isometric action means that each $g \in \Gamma$ acts on a Banach space E by an invertible linear isometry, with $g(hx) = (gh)x$. A vector y is *invariant* if $gy = y$ for every $g \in \Gamma$. We write gx instead of $\pi(g)x$ except where the representation must be made explicit.

For a finitely supported probability distribution μ on Γ, write

$$\mu x = \sum_g \mu(g)\, gx.$$

This is a contraction and fixes every invariant vector. Its *radius* is the maximum word length of a group element in its support, relative to a fixed finite symmetric generating set S.

We use uniform convexity in its closed-unit-ball form: there is a nondecreasing function $\delta : (0, 2] \to (0, 1/2]$ such that

$$\|a\|, \|b\| \le 1, \quad \|a - b\| \ge \varepsilon \quad \Longrightarrow \quad \left\| \frac{a+b}{2} \right\| \le 1 - \delta(\varepsilon). \tag{1.1}$$

This is equivalent to the usual unit-sphere definition; an elementary conversion is given in Section B. All constants below may depend on this common lower bound for the modulus, but not on the action.

Theorem 1.1 (One finite average moves closer to invariants)**.** *There are a finitely supported probability distribution μ on Γ and a number $\theta < 1$, depending only on q and δ, with the following property. For every linear isometric action on a Banach space satisfying* (1.1)*, every vector x, and every invariant vector y, there is an invariant vector y' such that*

$$\boxed{\|Ax - y'\| \le \theta \|x - y\|, \qquad A = \pi(\mu).} \tag{1.2}$$

The distribution μ is fixed before π, x, y are chosen. In particular, Γ has Banach property (T_E) for every uniformly convex E.

This is a rewrite of the infinite-group argument in [1, Section 2, Theorem 6]. Its main averaging products and interpolation are unchanged. We make the last step explicit instead of invoking a relative property-(T) synthesis theorem. We also make a useful finite-stage observation: both six-factor products are exactly invariant under an increasing finite upper-triangular subgroup. This lets us avoid introducing any limiting operator.

The algebraic input: boundedly many triangular factors

Let $U, V < \Gamma$ be the upper- and lower-unitriangular subgroups. We use the following consequence of Nica's theorem [3, Theorem 2]:

$$\boxed{\text{Every } g \in \Gamma \text{ is a product of at most } K \text{ elements of } U \cup V.} \tag{1.3}$$

One may take $K = 41$: Nica proves that 41 elementary matrices suffice in dimension three, and each elementary matrix belongs to U or V. The coefficients in these factors can have arbitrarily large degree. *Bounded generation does not mean bounded word length in S.* We do not reprove this algebraic theorem.

The averaging input: two short averages

The main calculation will establish the following statement.

Proposition 1.2 (Short averages with uniform subgroup error)**.** *For every $\tau > 0$ there are finitely supported probability distributions μ_+, μ_- on Γ such that, in every action under consideration, $B_+ = \pi(\mu_+)$ and $B_- = \pi(\mu_-)$ satisfy*

$$\|hB_+z - B_+z\| \le 2\tau \,\|z\| \qquad (h \in U), \tag{1.4}$$

$$\|hB_-z - B_-z\| \le 2\tau \,\|z\| \qquad (h \in V). \tag{1.5}$$

Their supports depend only on q, δ, τ, not on the representation.

The quantifier is important: the same error bound holds for *every* element of the entire triangular subgroup, not just its short words. Sections 4 to 7 prove this proposition using finite averages and finite telescoping sums.

We also use one elementary geometric fact.

Lemma 1.3 (A small orbit contains a nearby invariant vector in its convex hull)**.** *Suppose a group acts linearly and isometrically on a uniformly convex Banach space, and*

$$\|gx - x\| \le r \qquad \text{for every } g.$$

Then there is an invariant vector y_* with $\|x - y_*\| \le r$. One may choose y_* to be the unique vector of least norm in the closed convex hull of the orbit of x.

The lemma uses no assumption of property (T). Its proof, from uniform convexity and completeness, is in Section 3.

2 The entire local-to-global step: two cases

Assume Proposition 1.2 and Lemma 1.3 for the moment. Fix

$$\varepsilon = \frac{1}{16K}, \qquad 0 < \tau \le \varepsilon, \qquad \eta = \delta(\varepsilon), \tag{2.1}$$

and choose B_+, B_- from Proposition 1.2. Define the single averaging operator

$$\boxed{A = \frac{\mathrm{Id} + B_+ + B_-}{3}, \qquad \theta = 1 - \frac{2\eta}{3} < 1.} \tag{2.2}$$

This is a predetermined finite recipe. There is no choice of A depending on the vector.

Take x and an invariant y, and set $z = x - y$. If $z = 0$, take $y' = y$. Otherwise there are precisely two cases.

Case I: one short average moves z noticeably

Suppose, for example,

$$\|B_+ z - z\| \ge \varepsilon \|z\| .$$

Both z and $B_+ z$ have norm at most $\|z\|$. Uniform convexity gives

$$\left\| \frac{z + B_+ z}{2} \right\| \le (1 - \eta) \|z\| .$$

The remaining term $B_- z$ has norm at most $\|z\|$, so

$$\begin{aligned} \|Az\| &\le \frac{2}{3} \left\| \frac{z + B_+ z}{2} \right\| + \frac{1}{3} \|B_- z\| \\ &\le \left(1 - \frac{2\eta}{3}\right) \|z\| . \end{aligned}$$

Since $Ay = y$, this is

$$\boxed{\|Ax - y\| \le \theta \|x - y\| .}$$

The same proof works if B_- is the average that moves z noticeably. **In this case, keep the original invariant vector: $y' = y$.**

Case II: both short averages barely move z

Now suppose

$$\|B_+ z - z\| < \varepsilon \|z\| , \qquad \|B_- z - z\| < \varepsilon \|z\| .$$

For every $h \in U$, Proposition 1.2 and the triangle inequality give

$$\begin{aligned} \|hz - z\| &\le \|h(z - B_+ z)\| + \|hB_+ z - B_+ z\| + \|B_+ z - z\| \\ &\le 2(\varepsilon + \tau) \|z\| . \end{aligned}$$

Using B_- gives the same bound for every $h \in V$.

Write an arbitrary $g \in \Gamma$ as a product of at most K elements of $U \cup V$. Telescoping along that product, with each prefix acting isometrically, yields

$$\boxed{\|gz - z\| \le 2K(\varepsilon + \tau)\,\|z\| \le \frac{1}{4}\,\|z\| \qquad (g \in \Gamma).} \tag{2.3}$$

Because y is invariant, $gx - x = gz - z$. Thus the *entire orbit of* x lies in the ball of radius $\|x - y\|/4$ around x. Lemma 1.3 supplies an invariant vector y_* with

$$\|x - y_*\| \le \frac{1}{4}\,\|x - y\|\,.$$

Finally A is a contraction and fixes y_*, so

$$\boxed{\|Ax - y_*\| = \|A(x - y_*)\| \le \frac{1}{4}\,\|x - y\|\,.}$$

Since $\eta \le 1/2$, we have $\theta \ge 2/3 > 1/4$. **In this case, use the new invariant vector:** $y' = y_*$**.** This proves Theorem 1.1 once the two ingredients are established.

The mechanism. An averaging step that moves the vector causes norm loss. If neither step moves it, the uniform subgroup estimates and bounded generation make its *whole orbit* small. This is why the argument is not circular: "almost invariant under a few generators" is never used as a substitute for being close to an invariant vector.

Why this is Banach property (T)

Let R bound the word lengths in the support of A, and put

$$e(x) = \max_{s \in S} \|sx - x\|\,.$$

A word of length r moves x by at most $re(x)$, so

$$\|x - Ax\| \le Re(x). \tag{2.4}$$

If there are no nonzero invariant vectors, Theorem 1.1, applied with $y = 0$, forces $y' = 0$ and gives $\|Ax\| \le \theta\,\|x\|$. Consequently

$$\boxed{e(x) \ge \frac{1 - \theta}{R}\,\|x\|\,.}$$

Thus unit vectors cannot be arbitrarily almost invariant.

The theorem also gives the usual quantitative statement in every representation. Let $D(x)$ denote the infimum of $\|x - y\|$ over invariant y. For each such y, the invariant y' constructed above satisfies

$$D(x) \le \|x - y'\| \le Re(x) + \theta\,\|x - y\|\,.$$

Taking the numerical infimum over y and rearranging gives

$$\boxed{D(x) \le \frac{R}{1 - \theta} \max_{s \in S} \|sx - x\|\,.} \tag{2.5}$$

No assertion that a closest invariant vector exists is used in this step. The invariant vector needed for the two-case proof was already constructed by the small-orbit lemma.

3 The elementary geometry of a small orbit

Here is the proof of Lemma 1.3. We include it so that no fixed-point theorem, reflexivity theorem, or metric-projection theorem is being hidden.

Let C be the closed convex hull of $\{gx : g \in \Gamma\}$. Explicitly, C consists of limits of finite convex combinations of orbit vectors. It is nonempty, bounded, closed, and convex, and every group element maps C onto itself. Since every orbit vector is within r of x, the triangle inequality shows

$$C \subseteq \{w : \|w - x\| \le r\}. \tag{3.1}$$

We show that C has a unique vector of least norm.

Put $a = \inf_{w \in C} \|w\|$, and choose $w_j \in C$ with $\|w_j\| \le a + 1/j$. If $a = 0$, these vectors converge to 0, which therefore lies in C and is its unique least-norm vector.

Suppose $a > 0$. Fix $0 < \beta \le a$. For large j, k, both norms are at most $a + t$, where $t > 0$ can be made arbitrarily small. If $\|w_j - w_k\| \ge \beta$, then, taking $t \le a$ and applying (1.1) after division by $a + t$,

$$\left\|\frac{w_j + w_k}{2}\right\| \le (a + t)\left(1 - \delta\left(\frac{\beta}{2a}\right)\right) < a$$

when t is sufficiently small. This contradicts the definition of a, since the midpoint lies in C. Thus (w_j) is Cauchy. Completeness gives a limit $w \in C$ with $\|w\| = a$.

Two distinct vectors of norm a in C would, by uniform convexity, have a midpoint of norm less than a. Hence w is unique. Every g maps C onto itself and preserves norms, so gw is also a least-norm vector in C. Uniqueness forces $gw = w$. Taking $y_* = w$ and using (3.1) proves the lemma.

This is the only place the proof needs completeness. The choice is a least-norm point of a *convex hull of an orbit*, not a closest point in the invariant subspace. The geometric argument is the small-orbit step used in Oppenheim's bounded-generation proof [4, proof of Theorem 4.3]; it is written out here in full.

4 The finite polynomial averages

We now prove Proposition 1.2. Fix an action and suppress π in formulas for averaging operators. All norm estimates are uniform over the actions satisfying (1.1).

For $d \ge 0$, let $\mathcal{P}_d$ be the additive group of polynomials of degree at most d, including zero. Define

$$C_{ij}(d) = \frac{1}{q^{d+1}} \sum_{p \in \mathcal{P}_d} e_{ij}(p). \tag{4.1}$$

Each $C_{ij}(d)$ is a finite average and thus a contraction. Adding a fixed $p \in \mathcal{P}_d$ permutes the summands, so

$$e_{ij}(p)C_{ij}(d) = C_{ij}(d) = C_{ij}(d)e_{ij}(p) \qquad (p \in \mathcal{P}_d). \tag{4.2}$$

In particular $C_{ij}(d)^2 = C_{ij}(d)$. This is just an identity of finite averages; no orthogonal projection is involved.

The calculation is organized inside a Heisenberg triple:

$$x(p) = e_{12}(p), \qquad y(q) = e_{23}(q), \qquad z(r) = e_{13}(r).$$

With $[a, b] = aba^{-1}b^{-1}$, the relevant relations are

$$[x(p), y(q)] = z(pq), \qquad z(r) \text{ commutes with all } x(p), y(q). \tag{4.3}$$

They follow by multiplying elementary matrices. Every triple of distinct indices gives the same relations, possibly with a minus sign in the central parameter. A minus sign never changes a uniform coefficient average.

Average a single polynomial coefficient at a time:

$$X_i = \mathrm{avg}_{c\in\mathbb{F}_q} x(ct^i), \quad Y_j = \mathrm{avg}_{c\in\mathbb{F}_q} y(ct^j), \quad Z_k = \mathrm{avg}_{c\in\mathbb{F}_q} z(ct^k).$$

Write $X^{\le r} = X_r \cdots X_0$, and similarly for Y, Z, with $Z^{\le -1} = \mathrm{Id}$. Then

$$C_{12}(r) = X^{\le r}, \quad C_{23}(r) = Y^{\le r}, \quad C_{13}(r) = Z^{\le r}.$$

Factors of the same letter commute, and all Z factors commute with all X and Y factors.

5 The Heisenberg contraction and degree pairing

Set

$$\eta_0 = \delta(1/2), \qquad \rho = 1 - \eta_0 \in [1/2, 1).$$

The local geometric input is the following lemma from the draft [1, Lemma 7].

Lemma 5.1 (One Heisenberg pair contracts)**.** *If $i + j = N$ and $Z_N\zeta = 0$, then*

$$\left\|X_iY_j\zeta\right\| \le \rho \left\|\zeta\right\|.$$

Proof. Put $\omega = Y_j\zeta$. Then $Z_N\omega = 0$, and every $y(ct^j)$ fixes ω. Since

$$\omega = \mathrm{avg}_{c\in\mathbb{F}_q}\big(\omega - z(ct^N)\omega\big),$$

there is $c_0 \in \mathbb{F}_q$ with $\left\|\omega - z(c_0t^N)\omega\right\| \ge \|\omega\|$. Let $a = x(t^i)$ and $b = y(c_0t^j)$, so $[a, b] = z(c_0t^N)$ and $b\omega = \omega$. The triangle inequality gives

$$\begin{aligned}\|\omega\| &\le \left\|\omega - aba^{-1}\omega\right\| \\ &\le \|\omega - a\omega\| + \left\|a\omega - aba^{-1}\omega\right\| \\ &= 2\|\omega - a\omega\|.\end{aligned}$$

For the last equality use $b\omega = \omega$ and the fact that a, b preserve norms. If $\omega \ne 0$, uniform convexity applied to ω and $a\omega$ therefore gives

$$\left\|\frac{\omega + a\omega}{2}\right\| \le \rho\|\omega\|.$$

But $X_ia = X_i$, so

$$\|X_i\omega\| = \left\|X_i\frac{\omega + a\omega}{2}\right\| \le \rho\|\omega\| \le \rho\|\zeta\|.$$

The case $\omega = 0$ is immediate. □

In words: if a central coefficient has mean zero, then two transverse coefficient averages must lose a definite amount of norm. We next repeat this contraction using the many decompositions $N = i + j$ of one degree.

Lemma 5.2 (Changing one central degree)**.** *For $r, s, N \ge 0$, put*

$$b_{r,s}(N) = \#\{(i, j) : 0 \le i \le r,\ 0 \le j \le s,\ i + j = N\}.$$

Then

$$\left\|X^{\le r}Z^{\le N-1}Y^{\le s} - X^{\le r}Z^{\le N}Y^{\le s}\right\| \le 2\rho^{b_{r,s}(N)}. \tag{5.1}$$

Proof. Let $D = Z^{\le N-1}(\mathrm{Id} - Z_N)$. The difference in (5.1) is

$$X_r \cdots X_0 Y_s \cdots Y_0 D.$$

We have $\|D\| \le 2$. Its range is preserved by all X_i, Y_j. On that range every central element $z(ct^k)$ with $k < N$ acts as the identity, and Z_N acts as zero.

Process the pairs $i + j = N$ in decreasing order of i. Move the corresponding Y_j left until it is immediately after X_i. The Y factors commute with one another. Every $X_{i'}$ crossed satisfies $i' < i$, hence $i' + j < N$. For these crossings the group relation

$$x(at^{i'})y(bt^j) = y(bt^j)x(at^{i'})z(abt^{i'+j})$$

shows, after averaging a, b, that $X_{i'}Y_j$ and $Y_jX_{i'}$ *agree on the range of* D. Thus the reordering is legitimate there.

There are now $b_{r,s}(N)$ adjacent matched pairs X_iY_j. Each contracts by ρ on $\ker Z_N$, by Lemma 5.1. All intervening factors preserve that subspace and are contractions. Multiplying their bounds, and then using $\|D\| \le 2$, proves the claim. □

The proof uses commutator identities for *group elements*, followed by averaging. It does not treat a commutator of noninvertible averaging operators as an average of group commutators. This is the precise reordering formulation also used in the companion note [2, Lemma 4.1].

We will also use the exact identity

$$X^{\le r}Z^{\le m}Y^{\le s} = Y^{\le s}Z^{\le m}X^{\le r} \qquad \text{if } r + s \le m. \tag{5.2}$$

Indeed, commuting $x(p)$ past $y(q)$ changes the central parameter by $\pm pq$, whose degree is at most $r + s$. That change simply permutes the uniform central averaging set $\mathcal{P}_m$.

6 The six-factor interpolation

Keep the draft's two products [1, equations (18)–(19)]:

$$g_n = C_{12}(n)C_{13}(2n)C_{23}(2n)C_{21}(2n)C_{31}(2n)C_{32}(n), \tag{6.1}$$
$$h_n = C_{23}(n)C_{13}(2n)C_{12}(2n)C_{32}(2n)C_{31}(2n)C_{21}(n), \tag{6.2}$$

for $n \ge 1$. These symbols denote probability distributions as well as their averaging operators in the fixed representation.

Proposition 6.1 (The switching estimate)**.** *For every* $n \ge 1$,

$$\|g_n - h_{n+1}\|,\ \|h_n - g_{n+1}\| \le (8n + 12)\rho^{n-1} \le 20n\rho^{n-1}. \tag{6.3}$$

Proof. Abbreviate $C_{ij}(d)$ by $[ij]_d$. The complete interpolation is

$$\begin{aligned}
g_n &= [12]_n[13]_{2n}[23]_{2n}[21]_{2n}[31]_{2n}[32]_n \\
&\overset{1}{\approx} [12]_n[13]_{2n+2}[23]_{2n}[21]_{2n}[31]_{2n}[32]_n \\
&\overset{2}{\approx} [12]_n[13]_{2n+2}[23]_{2n}[21]_{2n}[31]_{2n+2}[32]_n \\
&\overset{3}{\approx} [12]_n[13]_{2n+2}[23]_{n+1}[21]_{2n}[31]_{2n+2}[32]_n \\
&\overset{4}{\approx} [12]_n[13]_{2n+2}[23]_{n+1}[21]_{n+1}[31]_{2n+2}[32]_n \\
&= [23]_{n+1}[13]_{2n+2}[12]_n[21]_{n+1}[31]_{2n+2}[32]_n \\
&= [23]_{n+1}[13]_{2n+2}[12]_n[32]_n[31]_{2n+2}[21]_{n+1} \\
&\overset{5}{\approx} [23]_{n+1}[13]_{2n+2}[12]_{2n+2}[32]_n[31]_{2n+2}[21]_{n+1} \\
&\overset{6}{\approx} [23]_{n+1}[13]_{2n+2}[12]_{2n+2}[32]_{2n+2}[31]_{2n+2}[21]_{n+1} = h_{n+1}.
\end{aligned}$$

The two equalities use (5.2), since $n + (n+1) \le 2n+2$. Each approximate step changes the middle factor of one Heisenberg triple. Apply Lemma 5.2 once for each degree crossed; all factors outside that triple are contractions.

Here is the complete error accounting. The triple is ordered as (noncentral, central, noncentral). The quantities r, s are the cutoffs of the two noncentral factors.

Step	Triple	(r,s)	Degrees N crossed	$b_{r,s}(N) \ge$
1	$(12,13,23)$	$(n,2n)$	$2n+1,\ 2n+2$	$n-1$
2	$(21,31,32)$	$(2n,n)$	$2n+1,\ 2n+2$	$n-1$
3	$(13,23,21)$	$(2n+2,2n)$	$n+2,\dots,2n$	$n+3$
4	$(23,21,31)$	$(n+1,2n+2)$	$n+2,\dots,2n$	$n+2$
5	$(13,12,32)$	$(2n+2,n)$	$n+1,\dots,2n+2$	$n+1$
6	$(12,32,31)$	$(2n+2,2n+2)$	$n+1,\dots,2n+2$	$n+2$

These counts follow from

$$b_{r,s}(N) = \max\{0, \min(r,N) - \max(0, N-s) + 1\}.$$

When $n = 1$, steps 3 and 4 are empty. There are

$$2 + 2 + (n-1) + (n-1) + (n+2) + (n+2) = 4n + 6$$

one-degree changes, each costing at most $2\rho^{n-1}$. This gives $(8n+12)\rho^{n-1}$.

For the other inequality, let J reverse the three coordinates and use the group automorphism

$$\alpha(g) = J(g^{-1})^{\mathsf{T}}J.$$

It sends $e_{ij}(p)$ to $e_{4-j,4-i}(-p)$, and therefore interchanges g_n and h_n: the minus sign does not change any uniform coefficient average. Applying the first estimate to the representation $\pi \circ \alpha$ gives the second. Finally $8n + 12 \le 20n$ for $n \ge 1$. □

This is the main calculation. Its structure is simple even though there are six edits: change one central range at a time, pay an exponentially small error, and use two exact swaps when the central range is large enough to absorb all commutators.

7 Finite comparisons give the two short averages

Define

$$f_n = \begin{cases} g_n, & n \text{ odd}, \\ h_n, & n \text{ even}. \end{cases}$$

By Proposition 6.1, for every finite $M \geq N \geq 1$,

$$\|f_N - f_M\| \leq \sum_{n=N}^{M-1} 20n\rho^{n-1} \leq \tau_N, \qquad \tau_N = 20\rho^{N-1}\left(\frac{N}{1-\rho} + \frac{\rho}{(1-\rho)^2}\right). \tag{7.1}$$

The formula for τ_N is the tail of a scalar geometric-series calculation. Only the displayed *finite* comparison of two averaging operators is used; we never define an operator f_∞.

Exact invariance at a sufficiently large finite stage

For $n \geq 1$ let

$$U_n = \left\{ \begin{pmatrix} 1 & p & r \\ 0 & 1 & q \\ 0 & 0 & 1 \end{pmatrix} : \deg p, \deg q \leq n,\ \deg r \leq 2n \right\}. \tag{7.2}$$

Zero coefficients are allowed. This is a finite subgroup: multiplication only adds an extra product pq' to the $(1,3)$ coefficient, of degree at most $2n$. The groups U_n increase and exhaust U.

Lemma 7.1 (Both products are already invariant under U_n)**.** *For every $u \in U_n$,*

$$ug_n = g_n, \qquad uh_n = h_n.$$

These are exact identities of finitely supported probability distributions.

Proof. Consider the first three factors of g_n:

$$C_{12}(n)C_{13}(2n)C_{23}(2n) = \operatorname*{avg}_{\substack{p \in \mathcal{P}_n \\ r,q \in \mathcal{P}_{2n}}} x(p)z(r)y(q).$$

Left multiplication by $x(a)$ with $\deg a \leq n$ shifts p, and left multiplication by $z(c)$ with $\deg c \leq 2n$ shifts r. Both preserve this distribution. For $b \in \mathcal{P}_n$, use

$$y(b)x(p)z(r)y(q) = x(p)z(r - pb)y(q + b).$$

Since $\deg(pb) \leq 2n$, the change $(p, r, q) \mapsto (p, r - pb, q + b)$ is a bijection of the averaging set. Thus left multiplication by $y(b)$ also preserves the distribution. These three families generate U_n.

For h_n, the first three factors are $C_{23}(n)C_{13}(2n)C_{12}(2n)$. The same argument with x and y interchanged applies. Multiplying by the remaining three factors on the right preserves these left-invariance identities. □

This observation is why two short averages suffice, rather than six separate root-direction averages.

A single short average works uniformly for every upper element

Fix N and any $u \in U$. Choose a finite $M \geq N$ large enough that $u \in U_M$. By Lemma 7.1, $uf_M = f_M$. Hence, using (7.1),

$$\|uf_N - f_N\| = \|u(f_N - f_M) - (f_N - f_M)\| \leq 2\tau_N. \tag{7.3}$$

Crucially, M may depend on u, but the bound $2\tau_N$ does not. Thus (7.3) holds for *every* $u \in U$. This is a uniform estimate for the entire infinite subgroup, obtained by comparing finitely many operators at a time.

Choose N so that $\tau_N \leq \tau$ and let $\mu_+ = f_N$. Let $\iota(g) = (g^{-1})^{\mathsf{T}}$, which exchanges U and V, and define μ_- by applying ι to each group element in μ_+. The same estimate in the representation $\pi \circ \iota$ gives (1.5). Both distributions have finite support. This completes the proof of Proposition 1.2, and therefore of Theorem 1.1.

8 The finished recipe and the role of each ingredient

The actual recipe can now be written with no unspecified averaging operator. Take $K = 41$, $\varepsilon = 1/(16K)$, $\rho = 1 - \delta(1/2)$, and choose one integer N with

$$20\rho^{N-1}\left(\frac{N}{1-\rho} + \frac{\rho}{(1-\rho)^2}\right) \leq \varepsilon.$$

Use the six-factor word f_N above and its lower-triangular image $\iota_* f_N$. The required finite distribution is

$$\boxed{\mu = \frac{\mathbf{1}_e + f_N + \iota_* f_N}{3}, \qquad \theta = 1 - \frac{2}{3}\delta\left(\frac{1}{16K}\right).} \tag{8.1}$$

Here $\mathbf{1}_e$ is unit mass at the identity. No power of this distribution is needed to obtain the stated contraction.

For each x and invariant y, the two-case proof uses just two candidate invariant vectors: the given y, or the least-norm vector in the closed convex hull of the orbit of x. The first is used when a short average moves $x - y$; the second is used when bounded generation makes that orbit small.

There are three logically separate inputs. The Heisenberg calculation uses uniform convexity at separation $1/2$ to make changing large-degree averaging ranges cheap. The final norm-drop case uses uniform convexity at the smaller, fixed separation $1/(16K)$. The small-orbit lemma uses uniform convexity and completeness to obtain an *exactly* invariant vector. We therefore assert the theorem for uniformly convex Banach spaces, not merely for the single-scale norm hypothesis used elsewhere in the supplied notes.

The six-factor products and their interpolation are those of [1, Section 2]; the explicit degree bookkeeping and correct restricted reordering are also detailed in [2, Section 4]. The finite-stage U_n invariance, the use of finite comparisons in place of limiting operators, and the final single-average two-case packaging are the present reorganization. Nica supplies (1.3); the small-orbit synthesis follows the mechanism of [4, Theorem 4.3], with its geometric step proved above.

A A concrete finite generating set and radius bound

For an entirely explicit word metric, use

$$S = \{e_{ij}(a), e_{ij}(at) : i \neq j,\ a \in \mathbb{F}_q^{\times}\}.$$

This set is finite and symmetric. It generates every elementary matrix: for distinct i, j, k and $r \geq 2$,

$$e_{ij}(at^r) = [e_{ik}(at^{r-1}), e_{kj}(t)],$$

and $e_{ij}(p+p') = e_{ij}(p)e_{ij}(p')$. Nica's theorem then gives generation of Γ.

For a monomial of degree $r \geq 1$, its length is at most $w_r = 3 \cdot 2^{r-1} - 2$: start with $w_1 = 1$ and use $w_r \leq 2w_{r-1} + 2$. Thus every $e_{ij}(p)$ with $\deg p \leq D$ has length at most

$$1 + \sum_{r=1}^{D} (3 \cdot 2^{r-1} - 2) = 3 \cdot 2^D - 2D - 2 \leq 3 \cdot 2^D.$$

Each supporting word of f_N has six such factors of degree at most $2N$. The automorphism ι preserves S. Therefore the distribution (8.1) has radius at most

$$\boxed{R \leq 18 \cdot 4^N.}$$

The bound is deliberately crude. For any other fixed finite generating set, replace each element of S by a fixed word in that set.

B From unit-sphere to unit-ball convexity

Suppose the usual modulus gives midpoint deficit $d(\varepsilon) > 0$ for two unit vectors at distance at least ε. We verify (1.1) directly. Fix $0 < \varepsilon \leq 2$ and put

$$t = \min\{\varepsilon/4,\ d(\varepsilon/2)/2\}.$$

Let $\|a\|, \|b\| \leq 1$ and $\|a - b\| \geq \varepsilon$. If either norm is at most $1 - t$, then $\|(a+b)/2\| \leq 1 - t/2$ by the triangle inequality. Otherwise both vectors are nonzero; put $a' = a/\|a\|$ and $b' = b/\|b\|$. Then

$$\|a - a'\|, \|b - b'\| < t, \qquad \|a' - b'\| \geq \varepsilon - 2t \geq \varepsilon/2.$$

Consequently

$$\left\|\frac{a+b}{2}\right\| \leq \left\|\frac{a'+b'}{2}\right\| + t \leq 1 - d(\varepsilon/2) + t \leq 1 - \frac{1}{2}d(\varepsilon/2).$$

Thus a valid unit-ball modulus is

$$\delta(\varepsilon) = \min\{\varepsilon/8,\ d(\varepsilon/2)/4\}.$$

It is nondecreasing if d is, and is at most $1/2$, as assumed.

References

[1] *Some Banach Property (T) stuff*. Unpublished draft, June 28, 2026, Section 2 and Theorem 6. Supplied as `banach-property-t(7).pdf`.

[2] *A complete proof of Theorem 3: The three-move super-expanders over* $\mathbb{F}_2[t]/(t^\ell)$. Companion proof-completion note, September 8, 2026. Supplied as `super-expander-deets.pdf`. Only its averaging calculation is used here.

[3] Bogdan Nica. On bounded elementary generation for SL_n over polynomial rings. *Israel J. Math.* **225** (2018), 403–410. doi:10.1007/s11856-018-1666-4. arXiv:1901.00587. Theorem 2 and the explicit bound immediately following it.

[4] Izhar Oppenheim. Banach property (T) for $\mathrm{SL}_n(\mathbb{Z})$ and its applications. *Invent. Math.* **234** (2023), 893–930. doi:10.1007/s00222-023-01211-7. arXiv:2207.04407. Theorem 4.3, especially the small-orbit argument in its proof.